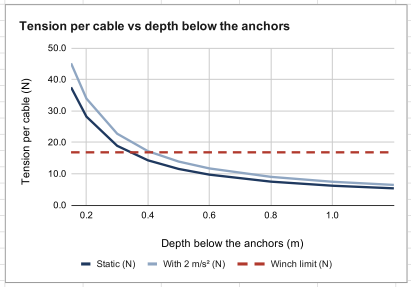
Cable tension

T = W / (4 sin φ), sin φ = d / √(d² + D_h²), claw centred

Input	Value	Unit	Note
Claw mass m_c	0.6	kg	weighed, with battery
Payload m_p	0.454	kg	1 lb test
Gravity g	9.81	m/s²	
Acceleration a	2	m/s²	design
Horizontal distance D_h	2.17	m	room centre to corner
Winch design force F	16.80	N	Winch pull tab
Weight W	10.34	N	W = (m_c + m_p) g

Depth d (m)	sin φ	Static (N)	With 2 m/s² (N)	Winch limit (N)
0.15	0.069	37.5	45.1	16.8
0.2	0.092	28.2	33.9	16.8
0.3	0.137	18.9	22.7	16.8
0.4	0.181	14.3	17.2	16.8
0.5	0.225	11.5	13.9	16.8
0.6	0.266	9.7	11.7	16.8
0.8	0.346	7.5	9.0	16.8
1	0.419	6.2	7.4	16.8
1.2	0.484	5.3	6.4	16.8



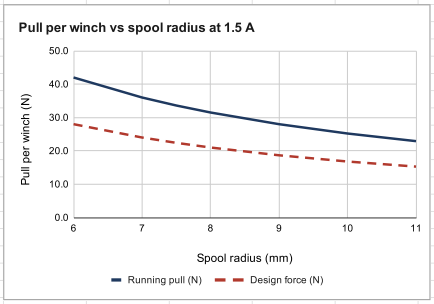
Winch pull

I = V_ref / (8 R_s), τ = k_r τ_h I / I_rated, T = τ / r, F = T_full / SF

Input	Value	Unit	Note
Holding torque τ_h	0.36	N·m	rated, at 1.5 A
Rated current I_rated	1.5	A	motor rating
Reference voltage V_ref	1.2	V	set for 1.5 A
Sense resistor R_s	0.1	Ω	A4988 carrier
Running / holding ratio k_r	0.7	-	assumed; pull test to measure
Skip margin SF	1.5	-	
Spool radius, empty r_e	7.5	mm	Ø15 core
Spool radius, full r_f	10	mm	line wound on

Result	Value	Unit	Formula
Driver current I	1.50 [1]	A	I = V_ref / (8 R_s)
Running torque τ	0.25	N·m	τ = k_r τ_h I / I_rated
Pull, empty spool	33.60	N	T = τ / r_e
Pull, full spool T_full	25.20	N	T_full = τ / r_f
Design force F	16.80	N	F = T_full / SF

Spool radius (mm)	Running pull (N)	Design force (N)
6	42.0	28.0
7	36.0	24.0
7.5	33.6	22.4
8	31.5	21.0
9	28.0	18.7
10	25.2	16.8
11	22.9	15.3



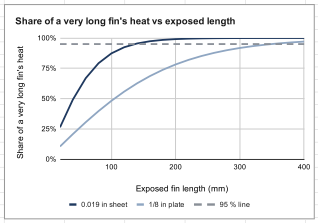
Heat fin

m = √(2h / (k t)), η = tanh(mL) / (mL), R_fin = 1 / (η h 2 w L) (Incropera, ch. 3)

Input	Value	Unit	Note
Conductivity k	160	W/m·K	aluminum
Film coefficient h	7	W/m²·K	still air + radiation, assumed
Fin width w	44	mm	
Thickness t, built	0.483	mm	0.019 in sheet
Thickness t, drawn	3.175	mm	1/8 in plate
Exposed length L	200	mm	as built

Result	Value	Unit	Formula
m, built	13.5	1/m	m = √(2h / (k t))
m, drawn	5.2	1/m	m = √(2h / (k t))
L_95, built	136.1	mm	tanh(m L_95) = 0.95
L_95, drawn	348.9	mm	tanh(m L_95) = 0.95
η at L, built	37%	-	η = tanh(mL) / (mL)
R_fin at L, built	22.1	K/W	R_fin = 1 / (η h 2 w L)
R_fin at L, drawn	10.9	K/W	R_fin = 1 / (η h 2 w L)

L (mm)	η, 0.019 in	G, 0.019 in (W/K)	η, 1/8 in	G, 1/8 in (W/K)	0.019 in sheet	1/8 in plate	95 % line
20	98%	0.012	100%	0.012	26%	10%	95%
40	91%	0.023	99%	0.024	49%	21%	95%
60	83%	0.031	97%	0.036	67%	30%	95%
80	74%	0.036	95%	0.047	79%	40%	95%
100	65%	0.040	92%	0.057	87%	48%	95%
120	57%	0.042	89%	0.065	92%	56%	95%
140	51%	0.044	85%	0.073	95%	63%	95%
160	45%	0.045	82%	0.080	97%	69%	95%
180	41%	0.045	78%	0.087	98%	74%	95%
200	37%	0.045	74%	0.092	99%	78%	95%
220	34%	0.046	71%	0.096	99%	82%	95%
240	31%	0.046	68%	0.100	100%	85%	95%
260	29%	0.046	64%	0.103	100%	88%	95%
280	27%	0.046	61%	0.106	100%	90%	95%
300	25%	0.046	58%	0.108	100%	92%	95%
320	23%	0.046	56%	0.109	100%	93%	95%
340	22%	0.046	53%	0.111	100%	95%	95%
360	21%	0.046	51%	0.112	100%	96%	95%
380	20%	0.046	48%	0.113	100%	96%	95%
400	19%	0.046	46%	0.114	100%	97%	95%



Motor temperature

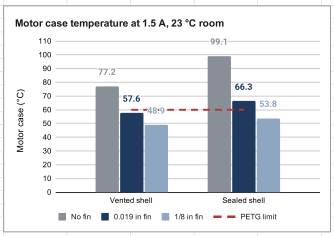
P = I² R, T = T_room + R_sa (P_shell + P_elec) + R_ma P_shell, P_shell = (P − R_sa P_elec / R_fin) / (1 + (R_sa + R_ma) / R_fin)

Input	Value	Unit	Note
Coil resistance R	1.8	Ω	assumed; to measure
Current I	1.50	A	Winch pull tab
Motor heat P	4.05	W	P = I² R
Electronics heat P_elec	1.5	W	estimate
Room air T_room	23	°C	near the ceiling
Contact resistance R_c	1	K/W	assumed
Fin path R_fin, 0.019 in	23.1	K/W	fin + contact
Fin path R_fin, 1/8 in	11.9	K/W	fin + contact

Enclosure	R_ma (K/W)	R_sa (K/W)	P_elec (W)	No fin (°C)	P_shell, 0.019 in (W)	0.019 in fin (°C)	P_shell, 1/8 in (W)	1/8 in fin (°C)
Open air	12	0	0	71.6	2.66	55.0	2.02	47.2
Vented shell	12	1	1.5	77.2	2.55	57.6	1.88	48.9
Sealed shell	14	3.5	1.5	99.1	2.17	66.3	1.46	53.8

Case	No fin	0.019 in fin	1/8 in fin	PETG limit
Vented shell	77.2	57.6	48.9	60
Sealed shell	99.1	66.3	53.8	60

PETG limit	60	°C	loaded printed parts



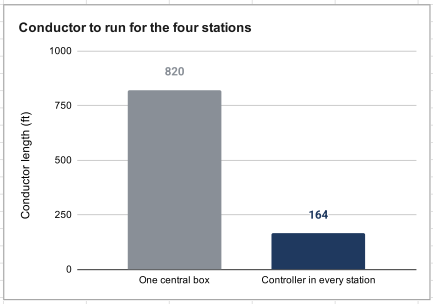
Wiring

Conductor length = n ΣL, voltage drop V = I R′ 2 L_max

Input	Value	Unit	Note
Route 1 (above the supply)	6	ft	measured
Route 2 (9 ft wall)	21	ft	measured
Route 3 (11 ft wall)	23	ft	measured
Route 4 (diagonal corner)	32	ft	measured
Conductors per corner, central box n_c	10	-	motor 4, switch 2, encoder 4
Conductors per corner, per station n_s	2	-	12 V, ground
Current per branch I	2	A	illustrative
Resistance R′, 16 AWG	4.016	Ω per 1000 ft	AWG table
Resistance R′, 18 AWG	6.385	Ω per 1000 ft	AWG table

Result	Value	Unit	Formula
Total route ΣL	82	ft	ΣL
Conductor, central box	820	ft	n_c ΣL
Conductor, per station	164	ft	n_s ΣL
Drop, longest run, 16 AWG	0.51	V	V = I R′ 2 L_max
Drop, longest run, 18 AWG	0.82	V	V = I R′ 2 L_max

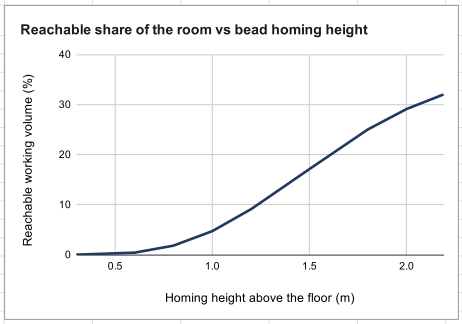
Architecture	Conductor length (ft)
One central box	820
Controller in every station	164



Bead homing

L_i ≥ L_i,home (beads homed with the claw at the room centre; 0.2 m wall and 0.4 m ceiling margins)

Home height (m) [2]	L_home (m)	Reachable volume (%)	
0.3	3.107	0	
0.6	2.892	0.4	
0.8	2.758	1.8	
1	2.633	4.7	
1.2	2.517	9.1	
1.5	2.364	17.1	
1.8	2.241	25	
2	2.178	29.1	
2.19	2.134	32	
Check			
Anchor height (m)	2.5932		anchors 0.05 m in from the corners
Anchor to centre, horizontal (m)	2.096		half diagonal less the inset
L_home at 1.0 m (m)	2.633		matches the table

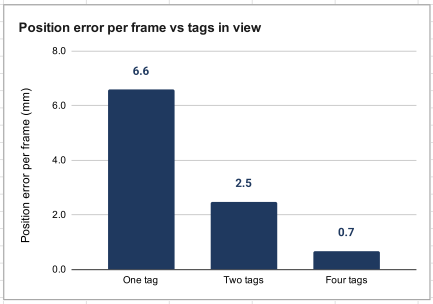


Floor board

δx = σ_θ L

Input	Value	Unit	Note
Camera height L	0.51	m	claw resting pose

Tags in view	σ_θ (mrad)	δx (mm)	Note
One tag	12.9	6.6	0.3 px corner noise, f = 520 px
Two tags	4.85	2.5	
Four tags	1.32	0.7	

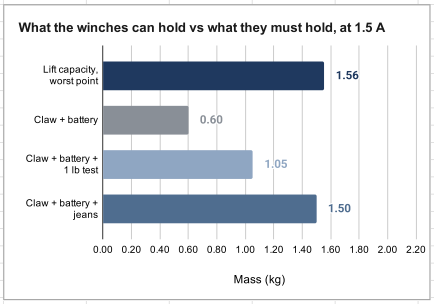


Lift budget

m_max = F / (k g)

Input	Value	Unit	Note
Lift capacity m_max	1.56	kg	F / (k g)
Worst tension per unit weight k	1.101	N/N	LP over the workspace, claw ≤ 1.6 m, incl. 2 m/s²
Printed parts, solid PETG	927	g	CAD
Print fill factor	0.55	-	assumed
Aluminum tube	198	g	CAD
Servo MG996R	55	g	datasheet
Battery	99	g	spec
Jeans	0.9	kg	design load
Claw, weighed (with battery)	600	g	weighed Oct 22
Result	**Value**	**Unit**	**Formula**
Claw without battery	0.50	kg	weighed − battery
Claw + battery	0.60	kg	
Claw + battery + jeans	1.50	kg	
Margin with jeans	0.06	kg	m_max − total
CAD estimate, with battery	0.86	kg	parts × fill + tube + servo + battery

Case	Mass (kg)
Lift capacity, worst point	1.56
Claw + battery	0.60
Claw + battery + 1 lb test	1.05
Claw + battery + jeans	1.50



Gear stress at servo stall

σ = F_t / (b m Y), F_t = 2τ / d, d = m z, SF = σ_allow / σ (Lewis)

Input	Value	Unit	Note
Stall torque τ (6.0 V)	1.079	N·m	MG996R, 11.0 kg·cm
Module m	1.5	mm	
Mesh efficiency η	0.95	-	assumed
Allowable stress σ_allow	22.5	MPa	PETG, half of 45 MPa

Gear	z	b (mm)	Y	τ (N·m)	d (mm)	F_t (N)	σ (MPa)	SF
Pinion (servo)	18	15.7	0.309	1.079	27	79.9	10.98	2.05
36T gear	36	15.7	0.3775	2.158	54	79.9	8.99	2.50
24T sync gear, one jaw	24	20	0.337	2.050	36	113.9	11.27	2.00

[1] Source: Allegro MicroSystems A4988 datasheet, current limit I_trip = V_ref / (8 R_s).

[2] Source: calculations/bead_reach.py in the RoomCleaner repository, run Oct 21, 2026 (grid of 60 x 50 x 50 points over the working volume).